

Nallabati Venkata Durga Sai Sarat Chandra

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SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive software architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech), Amrita Vishwa Vidyapeetham CGPA: 9.05 / 10.00	2024 – Present Amritapuri
Higher Secondary Education, Sri Chaitanya Vidhyanikethan CBSE (MPCEA) Percentage: 88.2%	2024 Vishakapatnam
Secondary School Education, Sri Chaitanya School SSC AP Marks: 570 / 600	2022 Rajam

SKILLS

Core Stack & Systems: Python, C, C++, SQL, Git, Linux, Ubuntu Server, ROS 2
AI, ML & SciML: Physics-Informed Machine Learning (PIML), Deep Learning, Model Optimization, Explainable AI
Robotics & Hardware: Sensor Fusion, Embedded Systems (Raspberry Pi, ESP32, Arduino), Communication Protocols, Control systems
Engineering & Simulation: Computational Fluid Dynamics (ANSYS Fluent), Thermodynamics, CAD (Fusion 360, AutoCAD)
Design & UI/UX Tools: Figma, Adobe Photoshop, Canva, UI/UX Wireframing, Responsive Layouts

PROJECTS

Evolution Edge: Self-Evolving Neural Bridge	05/2026
• Engineered a hybrid AI pipeline utilizing ROCm 6.x and ONNX Runtime to orchestrate low-latency inference on Ryzen AI NPUs with symbolic confidence-based escalation to AMD Instinct MI300X clusters. Implemented persistent on-device learning through knowledge distillation and PEFT-driven weight updates, bridging the gap between edge efficiency and cloud-scale intelligence.	
Physics-Informed Multi-Modal Industrial Diagnostic Framework	05/2026
• Engineered a Physics-Informed Machine Learning (PIML) framework combining mechanical heuristics and data-driven models to predict equipment failures. Optimized with SMOTE and Optuna for an 85.2% recall and integrated with SHAP for actionable transparency, the simulated system demonstrated a 68% reduction in maintenance costs, saving an estimated \$230,000 across a 2,000-machine fleet.	
AI-Driven Hemodynamic Digital Twin & Rheology Discovery	01/2026
• Developed a Physics-Informed Neural Network (PINN) framework to autonomously discover non-Newtonian blood rheology parameters from pulsatile CFD datasets of stenosed arteries. Engineered a high-accuracy hybrid pipeline integrating multi-fidelity data with boundary condition reconstruction, achieving <2% error in predicting complex hemodynamic flow fields.	
GSM-Based Smart Switch for Industrial Automation, Prof. Sumesh K.P	12/2025
• Developed a compact remote-automation node utilizing the GSM protocol for managing industrial loads in off-grid areas without internet access. Created a reliable asynchronous parser for SMS-based interrupts, ensuring over 99% uptime for remote hardware resets and power.	

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity, IEEE ICRM 2025	11/2025
• Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.	

ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.